SBT-DF204 COMPUTER FORENSICS CASE STUDIES

CASE STUDY 4: Investigating Rhino Possession Evidence

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Environment: Isolated Virtual Laboratory

Executive Summary

The examination identified a FAT16 USB image containing two allocated recipe text files and unallocated-space content from which seven JPEG candidates were carved. Network analysis identified FTP STOR requests for rhino1.jpg, rhino3.jpg (twice), and contraband.zip, with server responses indicating transfer completion. The two recovered rhino3.jpg data streams were byte-for-byte identical. An HTTP request for /~gnome/rhino.exe was followed by a 200 OK response, and the exported 145,920-byte object was identified as a Windows PE32 executable. The ZIP archive contains an encrypted entry named rhino2.jpg, but its contents were not verified.

Overall conclusion: the supplied artefacts support the existence of the recorded network requests, the recovered objects, and the observed file contents. They do not, by themselves, establish who operated the client, the legality or purpose of each file, original file provenance, or a direct relationship between the carved USB JPEGs and the FTP-recovered JPEGs. Confidence is high for the directly observed file types, hashes, and packet records; confidence is limited for attribution, intent, and any illegal-possession inference.

Objectives

The main objective of this investigation was to examine the supplied USB disk image and network traffic captures to identify, recover, and analyse digital artefacts relevant to the case.

The specific objectives were:

1. Preserve evidence integrity by creating working copies of the supplied evidence files and recording their SHA-256 hashes.
2. Examine the USB disk image to identify its filesystem, allocated files, unallocated space, and potentially recoverable data.
3. Recover and validate media files using file carving, file-type identification, metadata examination, and hash comparison.
4. Investigate potential hidden content in recovered media and document the results and limitations of any steganography analysis performed.
5. Analyse FTP traffic to identify file-upload requests, transfer directions, timestamps, server responses, and reconstructed files.
6. Examine the protected ZIP archive to identify its contents and encryption status, and document any limitations preventing content verification.
7. Analyse HTTP traffic to identify relevant requests and responses and reconstruct transferred objects, including the rhino.exe executable.
8. Construct a chronological timeline correlating material events across the USB image, FTP traffic, and HTTP traffic.
9. Draw evidence-based conclusions while distinguishing direct observations from inferences about file transfer, possession, intent, and user attribution.

2. Tools used

The following tools were used during the investigation, based on the commands and outputs recorded during analysis.

Tool	Purpose
sha256sum	Calculate SHA-256 hashes to document evidence integrity and compare recovered files.
file	Identify file types and validate recovered objects.
mmls	Examine the disk image for partition-layout information.
fsstat	Examine FAT16 filesystem properties and layout.
fls	List filesystem entries and search for deleted directory entries.
icat	Extract allocated files from the filesystem image.
tsk_recover	Attempt filesystem-based file recovery.
blkls	Extract unallocated filesystem content for further examination.
Foremost	Carve JPEG candidates from unallocated content.
ExifTool	Examine metadata and embedded comments in recovered files.

ImageMagick identify	Check image dimensions and basic image-format integrity indicators.
tshark	Analyse packet captures, FTP commands, HTTP requests, responses, and timestamps.
tcpflow	Reconstruct network data streams into files for examination.
unzip	Test the recovered ZIP archive; the encrypted entry could not be verified without a valid password.
zipinfo	Examine ZIP archive structure, entry names, sizes, compression, and encryption information.
strings	Extract readable strings from the executable for static inspection.
objdump	Examine PE executable headers, sections, and imported functions.
montage	Create contact sheets for visual examination of recovered images.

3. Methodology

The investigation followed a forensic workflow using the supplied training evidence and working copies.

3.1 Evidence preservation

The original files were kept in the evidence directory, with analysis performed on copies in the working directory. SHA-256 hashes were recorded for the supplied files and compared with working-copy hashes. The original evidence files were set to read-only to reduce the risk of accidental modification.

3.2 USB filesystem examination

The USB disk image, RHINOUSB.dd, was identified using file. The mmls and fsstat tools were used to examine its layout and filesystem properties, while fls was used to list allocated entries and search for deleted entries.

The allocated text files were extracted using icat. Filesystem-based recovery was attempted with tsk_recover, and blkls was used to extract unallocated filesystem content for additional analysis.

3.3 File carving and media validation

Foremost was used to carve JPEG candidates from the extracted unallocated content. The resulting files were examined using file, ExifTool, ImageMagick identifier, and SHA-256 hashing.

A contact sheet was generated with montage to visually inspect the recovered images. Hash comparisons were performed between the USB-carved JPEGs and FTP-reconstructed JPEGs to check for exact binary matches.

A limitation of the available results is that a completed steganography detection or hidden-content recovery result was not documented. Therefore, the report does not claim that hidden content was either confirmed or ruled out.

3.4 FTP traffic analysis

The rhino.log packet capture was examined using tshark to identify FTP control commands and server responses. The relevant FTP data streams were reconstructed using tcpflow.

The reconstructed objects were examined using file and sha256sum. This process identified transfers associated with rhino1.jpg, two uploads of rhino3.jpg, and contraband.zip. The hashes of the two recovered rhino3.jpg streams were compared to determine whether their contents were identical.

3.5 Protected archive examination

The reconstructed contraband.zip archive was examined using zipinfo and unzip -t. Its directory information identified an encrypted entry named rhino2.jpg.

Because the entry could not be decrypted and validated with the available results, its actual image content remains unverified. A matching uncompressed size between the archive entry and one USB-carved JPEG was treated only as a lead, not proof that the files were identical.

3.6 HTTP traffic and executable examination

The HTTP capture rhino3.log was examined using tshark to identify relevant requests, response codes, timestamps, and TCP streams. HTTP objects were exported for offline analysis.

The recovered rhino.exe object was checked with file and sha256sum. Its PE headers and imported functions were examined using objdump, and readable strings were extracted using strings.

The executable was not run, and no recorded host or service was contacted. Static-analysis observations were treated as leads rather than proof of malicious behaviour.

3.7 Timeline and interpretation

Relevant FTP and HTTP events were arranged chronologically and converted from the capture's displayed -0400 timezone to UTC. Packet numbers, TCP streams, file paths, hashes, and timestamps were recorded where available.

The final interpretation distinguishes technical observations from inferences. Network endpoints and file records alone cannot establish the identity of the person operating a device, the original provenance of a file, or unlawful possession or intent.

Evidence acquisition and integrity

The supplied originals were kept in the evidence directory and working copies were analysed from the working directory. The user reported that working-copy hashes matched the source evidence hashes and that the evidence files were made read-only. Hash values below are the recorded SHA-256 values.

Evidence ID	Original file	SHA-256
E01	evidence/rhino.log	64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2
E02	evidence/rhino2.log	41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde
E03	evidence/rhino3.log	7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46
E04	evidence/RHINOUSB.dd	ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65

Source and working paths: ~/icdfaDF/SBT-DF204-Case4/evidence/ and ~/icdfaDF/SBT-DF204-Case4/working/.

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/icdfaDF/SBT-DF204-Case4/{evidence,working,reports,screenshots}

┌──(kali㉿kali)-[~]
└─$ cd ~/icdfaDF/SBT-DF204-Case4

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ tree
.
├── evidence
├── reports
├── screenshots
└── working
```

sha256sum evidence/* | tee reports/original_hashes.txt

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
sha256sum evidence/* | tee reports/original_hashes.txt

41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde  evidence/rhino2.log
7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46  evidence/rhino3.log
64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2  evidence/rhino.log
ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65  evidence/RHINOUSB.dd
```

USB image and filesystem findings

The USB image was identified as a FAT16 filesystem. fsstat reported 512-byte sectors, 4,096-byte clusters, total sectors 0–506,847, FAT areas at sectors 1–248 and 249–496, data area at sectors 497–506,847, and cluster range 2–63,290. mmls produced no output; therefore, no partition table layout is asserted from that command. The available output is consistent with a FAT filesystem beginning at image offset zero.

The recursive file listing identified two allocated files: gumbo1.txt (metadata address 4) and gumbo2.txt (metadata address 6). They contained recipe text titled “SHRIMP AND TASSO GUMBO” and “SHRIMP AND ANDOUILLE SAUSAGE GUMBO.” The files were extracted with icat. No deleted entries were returned by fls -d, and tsk_recover reported zero files recovered. These results mean that those methods did not recover deleted files; they do not prove that no residual data exists in unallocated space.

Item	Observed result
USB format	FAT16; volume ID 0x4092d9d1
gumbo1.txt	2,815 bytes; SHA-256 ecb06cec73218f66d19cccfd1e45ab7144b02eb5344d9ede14e1c8428d76713c
gumbo2.txt	1,293 bytes; SHA-256 84cc7105cc6d663d230574d32256dd42bf96d7f077456e357ec9daacd59f87aa
Unallocated output	working/usb_unallocated.bin; 259,227,136 bytes; SHA-256

	67a45698872842cb0b2e69831eba8cde16a6594 9e7957dae62692ab4f7a4de3e

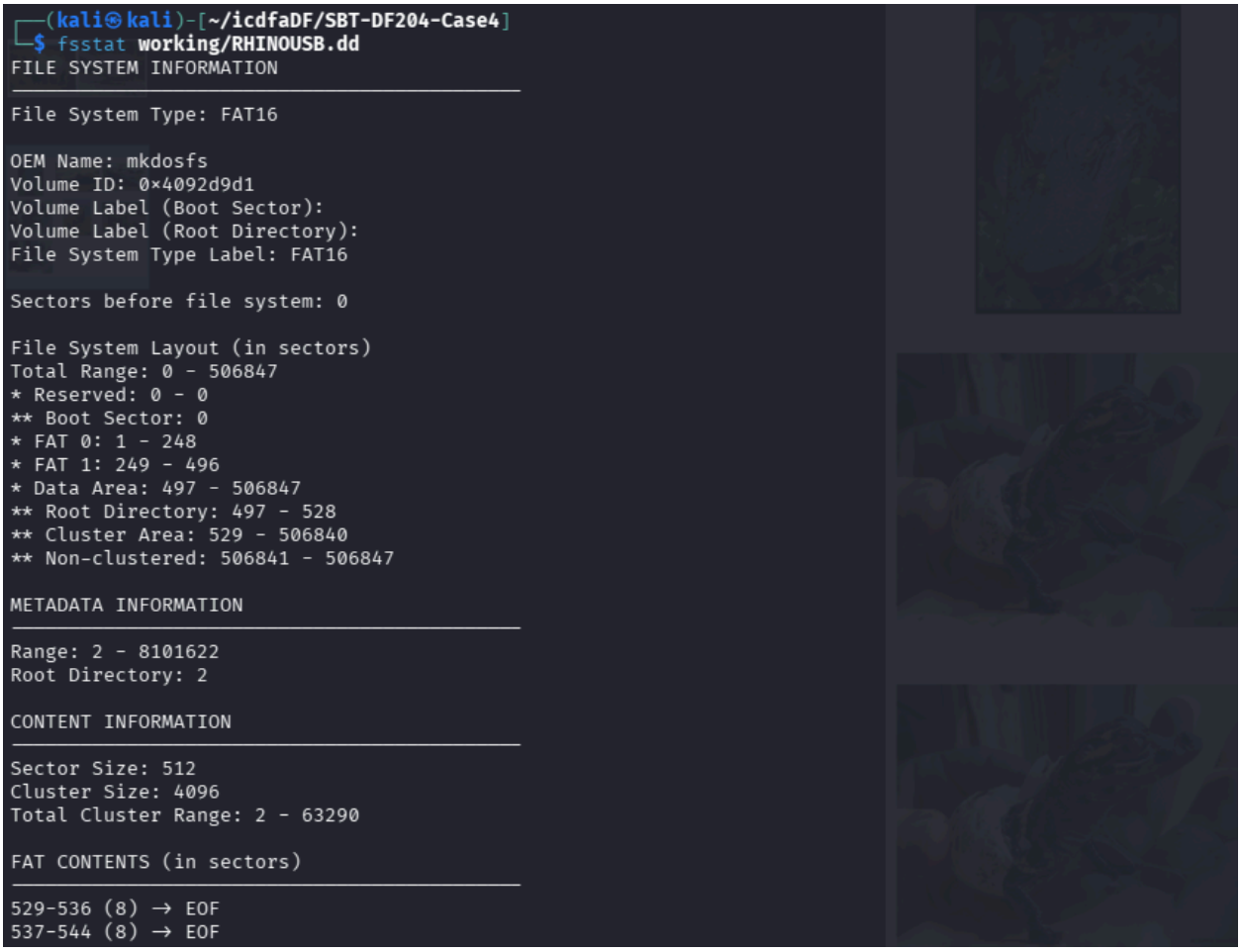
Commands used:

file working/RHINOUSB.dd



mmls working/RHINOUSB.dd

fsstat working/RHINOUSB.dd



fls -f fat -r -p working/RHINOUSB.dd

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ fls -f fat -r -p working/RHINOUSB.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:    $MBR
v/v 8101620:    $FAT1
v/v 8101621:    $FAT2
V/V 8101622:    $OrphanFiles

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
```

icat -f fat working/RHINOUSB.dd 4 > working/gumbo1.txt

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ icat -f fat working/RHINOUSB.dd 4 > working/gumbo1.txt

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
```

icat -f fat working/RHINOUSB.dd 6 > working/gumbo2.txt

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ icat -f fat working/RHINOUSB.dd 6 > working/gumbo2.txt

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
```

blkls -f fat working/RHINOUSB.dd > working/usb_unallocated.bin

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ blkls -f fat working/RHINOUSB.dd > working/usb_unallocated.bin

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
```

sha256sum working/gumbo1.txt working/gumbo2.txt working/usb_unallocated.bin

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ sha256sum working/gumbo1.txt working/gumbo2.txt working/usb_unallocated.bin
ecb06cec73218f66d19cccfd1e45ab7144b02eb5344d9ede14e1c8428d76713c  working/gumbo1.txt
84cc7105cc6d663d230574d32256dd42bf96d7f077456e357ec9daacd59f87aa  working/gumbo2.txt
67a45698872842cb0b2e69831eba8cde16a65949e7957dae62692ab4f7a4de3e  working/usb_unallocated.bin

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
```

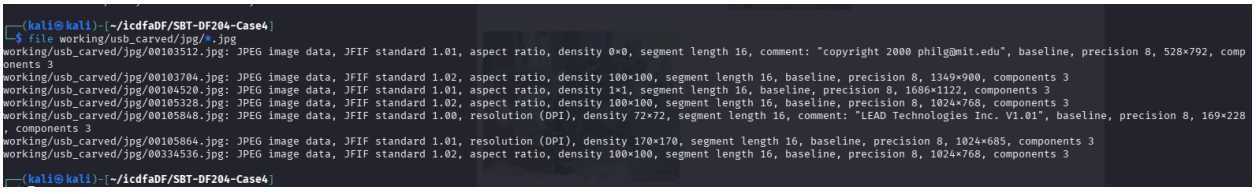
Carved USB media and metadata

Foremost was run against the extracted unallocated filesystem content and reported seven JPEG candidates. The carved offsets below are offsets within working/usb_unallocated.bin, not absolute offsets in RHINOUSB.dd. ImageMagick identified each as JPEG and the checked files ended with the JPEG EOI marker FF D9; this is a validation indicator, not proof that every image is wholly undamaged.

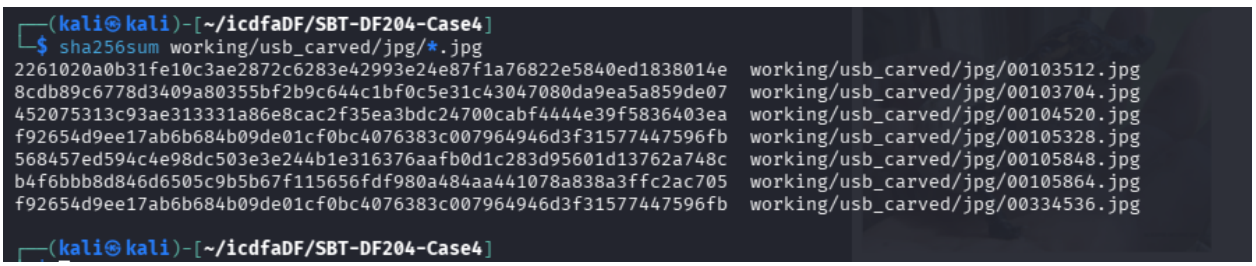
Carved file	Bytes	SHA-256 (shortened where noted)
00103512.jpg	95,814	2261020a0b31fe10c3ae2872c6283e42993e24e87f1a76822e5840ed1838014e
00103704.jpg	415,534	8cdb89c6778d3409a80355bf2b9c644c1bf0c5e31c43047080da9ea5a859de07
00104520.jpg	411,361	452075313c93ae313331a86e8cac2f35ea3bdc24700cabf4444e39f5836403ea
00105328.jpg	264,600	f92654d9ee17ab6b684b09de01cf0bc4076383c007964946d3f31577447596fb
00105848.jpg	6,809	568457ed594c4e98dc503e3e244b1e316376aafb0d1c283d95601d13762a748c
00105864.jpg	230,665	b4f6bbb8d846d6505c9b5b67f115656fdf980a484aa441078a838a3ffc2ac705
00334536.jpg	264,600	f92654d9ee17ab6b684b09de01cf0bc4076383c007964946d3f31577447596fb

The carved contact sheet visually showed wildlife images including rhinoceroses, crocodilians, and a hatchling reptile. Files 00105328.jpg and 00334536.jpg had identical hashes, establishing that they are duplicate byte sequences. No DateTimeOriginal/CreateDate/ModifyDate values were found for the carved JPEGs in the reported ExifTool review. Two files contained comments: 00105848.jpg had “LEAD Technologies Inc. V1.01” and 00103512.jpg had “copyright 2000 philg@mit.edu.” These comments do not establish authorship or the date the files were placed on the USB.

file working/usb_carved/jpg/*.jpg



sha256sum working/usb_carved/jpg/*.jpg



exiftool -r working/usb_carved/jpg/

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ exiftool -r working/usb_carved/jpg/
======== working/usb_carved/jpg/00104520.jpg
ExifTool Version Number         : 13.55
File Name                       : 00104520.jpg
Directory                       : working/usb_carved/jpg
File Size                       : 411 kB
File Modification Date/Time     : 2026:10:26 17:57:43-04:00
File Access Date/Time           : 2026:10:26 17:58:48-04:00
File Inode Change Date/Time     : 2026:10:26 17:57:43-04:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 1
Y Resolution                    : 1
Image Width                     : 1686
Image Height                    : 1122
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 1686×1122
Megapixels                      : 1.9
======== working/usb_carved/jpg/00105328.jpg
ExifTool Version Number         : 13.55
File Name                       : 00105328.jpg
Directory                       : working/usb_carved/jpg
File Size                       : 265 kB
File Modification Date/Time     : 2026:10:26 17:57:43-04:00
```

identify working/usb_carved/jpg/*.jpg

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ identify working/usb_carved/jpg/*.jpg
working/usb_carved/jpg/00103512.jpg JPEG 528×792 528×792+0+0 8-bit sRGB 95814B 0.000u 0:00.000
working/usb_carved/jpg/00103704.jpg JPEG 1349×900 1349×900+0+0 8-bit sRGB 415534B 0.000u 0:00.000
working/usb_carved/jpg/00104520.jpg JPEG 1686×1122 1686×1122+0+0 8-bit sRGB 411361B 0.000u 0:00.000
working/usb_carved/jpg/00105328.jpg JPEG 1024×768 1024×768+0+0 8-bit sRGB 264600B 0.000u 0:00.000
working/usb_carved/jpg/00105848.jpg JPEG 169×228 169×228+0+0 8-bit sRGB 6809B 0.000u 0:00.000
working/usb_carved/jpg/00105864.jpg JPEG 1024×685 1024×685+0+0 8-bit sRGB 230665B 0.000u 0:00.000
working/usb_carved/jpg/00334536.jpg JPEG 1024×768 1024×768+0+0 8-bit sRGB 264600B 0.000u 0:00.000

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
```

7. FTP traffic and reconstructed objects

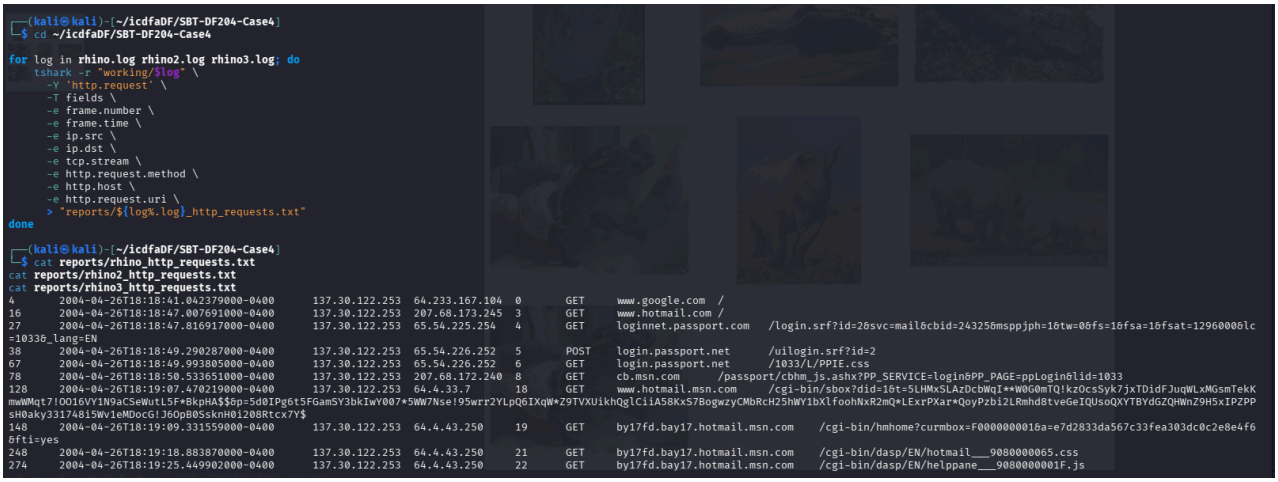
The FTP control traffic in rhino.log recorded STOR requests from client 137.30.122.253 to server 137.30.120.40. The relevant FTP data streams were reconstructed with tcpflow. FTP credentials are intentionally omitted from this report and should be redacted from screenshots.

Evidence ID	Time shown in capture (-0400)	UTC normalized	Observed FTP event

E05	2004-04-26 18:21:49.954	2004-04-26 22:21:49.954Z	STOR rhino1.jpg; server subsequently returned 150 and 226.
E06	2004-04-26 18:22:08.905	2004-04-26 22:22:08.905Z	STOR rhino3.jpg; server returned 150 and 226, with an ASCII-mode warning about bare linefeeds.
E07	2004-04-26 18:22:16.091	2004-04-26 22:22:16.091Z	Second STOR rhino3.jpg; binary-mode transfer; server returned 150 and 226.
E08	2004-04-26 18:26:46.671	2004-04-26 22:26:46.671Z	STOR contraband.zip; server returned 150 and 226.

The recovered FTP data objects were: rhino1.jpg, 65,703 bytes, SHA-256 0f30a8777d19a4fe4b2fafd5154b6c4dfafb673202db29971e876e978da4e05d; two rhino3.jpg streams, each 96,899 bytes, SHA-256 a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5; and contraband.zip, 230,566 bytes, SHA-256 b7091a0b4295a561bf76969073ede126a5ca75899bb115447e67272f95d5a29f. The two rhino3.jpg streams match exactly by SHA-256. Recovered JPEGs were identified as valid JPEG-format files, with dimensions 501×360 for rhino1.jpg and 720×597 for rhino3.jpg.

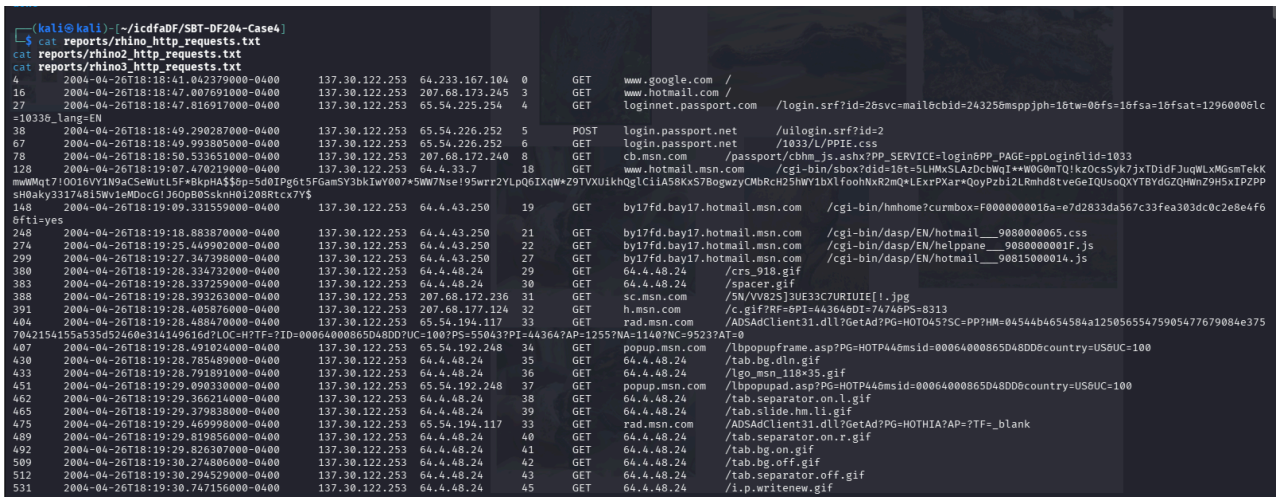
```
tshark -r working/rhino.log -Y 'ftp.request.command || ftp.response.code' -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e ftp.request.command -e ftp.response.code > reports/rhino_ftp_commands.txt
tcpflow -r working/rhino.log -o working/ftp_recovered
file working/ftp_recovered/*
sha256sum working/ftp_recovered/*
```



```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ cd ~/icdfaDF/SBT-DF204-Case4

for log in rhino.log rhino2.log rhino3.log; do
    tshark -r "working/$log" \
      -Y 'http.request' \
      -T fields \
      -e frame.number \
      -e frame.time \
      -e ip.src \
      -e ip.dst \
      -e tcp.stream \
      -e http.request.method \
      -e http.host \
      -e http.request.uri \
      > "reports/${log%.log}_http_requests.txt"
done

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ cat reports/rhino_http_requests.txt
cat reports/rhino2_http_requests.txt
cat reports/rhino3_http_requests.txt
4       2004-04-26T18:18:41.042379000-0400      137.30.122.253  64.233.167.104  0       GET     www.google.com  /
16      2004-04-26T18:18:47.007691000-0400      137.30.122.253  207.68.173.245  3       GET     www.hotmail.com /
27      2004-04-26T18:18:47.816917000-0400      137.30.122.253  65.54.225.254   4       GET     loginnet.passport.com   /login.srf?id=2&svc=mail&cbid=24325&msppjph=1&tw=0&fs=1&fsa=1&fsat=1296000&lc
=1033&_lang=EN
38      2004-04-26T18:18:49.290287000-0400      137.30.122.253  65.54.226.252   5       POST    login.passport.net      /uilogin.srf?id=2
67      2004-04-26T18:18:49.993805000-0400      137.30.122.253  65.54.226.252   6       GET     login.passport.net      /1033/L/PPIE.css
78      2004-04-26T18:18:50.533651000-0400      137.30.122.253  207.68.172.240  8       GET     cb.msn.com      /passport/cbhm_js.ashx?PP_SERVICE=login&PP_PAGE=ppLogin&lid=1033
128     2004-04-26T18:19:07.470219000-0400      137.30.122.253  64.4.33.7       18      GET     www.hotmail.msn.com     /cgi-bin/sbox?did=1&t=5LHMxSLAzDcbWqI**W0G0mTQ!kzOcsSyk7jxTDidFJuqWLxMGsmTekK
mwWMqt7!OO16VY1N9aCSeWutL5F*BkpHA$$0p=5d0IPg6t5FGamSY3bkIwY007*5WW7Nse!95wrr2YLpQ6IXqW*Z9TVXUikhQglCiiA58KxS7BogwzyCMbRcH25hWY1bXlfoohNxR2mQ*LExrPXar*QoyPzbi2LRmhd8tveGeIQUsoQXYTBYdGZQHWnZ9H5xIPZPP
sH0aky33174815Wv1eMDocG!J6OpB0SsknH0i208Rtcx7Y$
148     2004-04-26T18:19:09.331559000-0400      137.30.122.253  64.4.43.250     19      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/hmhome?curmbox=F0000000018a=e7d2833da567c33fea303dc0c2e8e4f6
8fti=yes
248     2004-04-26T18:19:18.883870000-0400      137.30.122.253  64.4.43.250     21      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/dasp/EN/hotmail___9080000065.css
274     2004-04-26T18:19:25.449902000-0400      137.30.122.253  64.4.43.250     22      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/dasp/EN/helppane___9080000001F.js
```



```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ cat reports/rhino_http_requests.txt
cat reports/rhino2_http_requests.txt
cat reports/rhino3_http_requests.txt
4       2004-04-26T18:18:41.042379000-0400      137.30.122.253  64.233.167.104  0       GET     www.google.com  /
16      2004-04-26T18:18:47.007691000-0400      137.30.122.253  207.68.173.245  3       GET     www.hotmail.com /
27      2004-04-26T18:18:47.816917000-0400      137.30.122.253  65.54.225.254   4       GET     loginnet.passport.com   /login.srf?id=2&svc=mail&cbid=24325&msppjph=1&tw=0&fs=1&fsa=1&fsat=1296000&lc
=1033&_lang=EN
38      2004-04-26T18:18:49.290287000-0400      137.30.122.253  65.54.226.252   5       POST    login.passport.net      /uilogin.srf?id=2
67      2004-04-26T18:18:49.993805000-0400      137.30.122.253  65.54.226.252   6       GET     login.passport.net      /1033/L/PPIE.css
78      2004-04-26T18:18:50.533651000-0400      137.30.122.253  207.68.172.240  8       GET     cb.msn.com      /passport/cbhm_js.ashx?PP_SERVICE=login&PP_PAGE=ppLogin&lid=1033
128     2004-04-26T18:19:07.470219000-0400      137.30.122.253  64.4.33.7       18      GET     www.hotmail.msn.com     /cgi-bin/sbox?did=1&t=5LHMxSLAzDcbWqI**W0G0mTQ!kzOcsSyk7jxTDidFJuqWLxMGsmTekK
mwWMqt7!OO16VY1N9aCSeWutL5F*BkpHA$$0p=5d0IPg6t5FGamSY3bkIwY007*5WW7Nse!95wrr2YLpQ6IXqW*Z9TVXUikhQglCiiA58KxS7BogwzyCMbRcH25hWY1bXlfoohNxR2mQ*LExrPXar*QoyPzbi2LRmhd8tveGeIQUsoQXYTBYdGZQHWnZ9H5xIPZPP
sH0aky33174815Wv1eMDocG!J6OpB0SsknH0i208Rtcx7Y$
148     2004-04-26T18:19:09.331559000-0400      137.30.122.253  64.4.43.250     19      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/hmhome?curmbox=F0000000018a=e7d2833da567c33fea303dc0c2e8e4f6
8fti=yes
248     2004-04-26T18:19:18.883870000-0400      137.30.122.253  64.4.43.250     21      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/dasp/EN/hotmail___9080000065.css
274     2004-04-26T18:19:25.449902000-0400      137.30.122.253  64.4.43.250     22      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/dasp/EN/helppane___9080000001F.js
299     2004-04-26T18:19:27.347398000-0400      137.30.122.253  64.4.43.250     27      GET     by17fd.bay17.hotmail.msn.com    /cgi-bin/dasp/EN/hotmail___90815000014.js
380     2004-04-26T18:19:28.334732000-0400      137.30.122.253  64.4.48.24      29      GET     64.4.48.24      /crs_918.gif
383     2004-04-26T18:19:28.337259000-0400      137.30.122.253  64.4.48.24      30      GET     64.4.48.24      /spacer.gif
388     2004-04-26T18:19:28.393263000-0400      137.30.122.253  207.68.172.236  31      GET     sc.msn.com      /5N/VV82S]3UE33C7URIUIE[!.jpg
391     2004-04-26T18:19:28.405876000-0400      137.30.122.253  207.68.177.124  32      GET     h.msn.com       /c.gif?RF=6PI=44364&DI=7474&PS=8313
404     2004-04-26T18:19:28.488470000-0400      137.30.122.253  65.54.194.117   33      GET     rad.msn.com     /ADSAdClient31.dll?GetAd?PG=HOTO45?SC=PP?HM=04544b4654584a1250565547590547679084e375
7042154155a535d52460e314149616d?LOC=H?TF=?ID=00064000865D48DD?UC=100?PS=5504&3?PI=44364?AP=1255?NA=1140?NC=9523?AT=0
407     2004-04-26T18:19:28.491024000-0400      137.30.122.253  65.54.192.248   34      GET     popup.msn.com   /lbpopupframe.asp?PG=HOTP44&msid=00064000865D48DD&country=US&UC=100
430     2004-04-26T18:19:28.785489000-0400      137.30.122.253  64.4.48.24      35      GET     64.4.48.24      /tab.bg.dln.gif
433     2004-04-26T18:19:28.791891000-0400      137.30.122.253  64.4.48.24      36      GET     64.4.48.24      /go_msn_110x35.gif
451     2004-04-26T18:19:29.090330000-0400      137.30.122.253  65.54.192.248   37      GET     popup.msn.com   /lbpopupad.asp?PG=HOTP44&msid=00064000865D48DD&country=US&UC=100
462     2004-04-26T18:19:29.366214000-0400      137.30.122.253  64.4.48.24      38      GET     64.4.48.24      /tab.separator.on.l.gif
465     2004-04-26T18:19:29.379838000-0400      137.30.122.253  64.4.48.24      39      GET     64.4.48.24      /tab.slide.hm.li.gif
475     2004-04-26T18:19:29.469998000-0400      137.30.122.253  65.54.194.117   33      GET     rad.msn.com     /ADSAdClient31.dll?GetAd?PG=HOTHIA?AP=?TF=_blank
489     2004-04-26T18:19:29.819856000-0400      137.30.122.253  64.4.48.24      40      GET     64.4.48.24      /tab.separator.on.r.gif
492     2004-04-26T18:19:29.826307000-0400      137.30.122.253  64.4.48.24      41      GET     64.4.48.24      /tab.bg.on.gif
509     2004-04-26T18:19:30.274806000-0400      137.30.122.253  64.4.48.24      42      GET     64.4.48.24      /tab.bg.off.gif
512     2004-04-26T18:19:30.294529000-0400      137.30.122.253  64.4.48.24      43      GET     64.4.48.24      /tab.separator.off.gif
531     2004-04-26T18:19:30.747156000-0400      137.30.122.253  64.4.48.24      45      GET     64.4.48.24      /i.p.writenew.gif
```

8. Protected ZIP archive

The recovered contraband.zip file was recognized as a ZIP archive. zipinfo reported a single entry named rhino2.jpg, encrypted, using Deflate compression. The entry had a reported uncompressed size of 230,665 bytes and CRC32 936ebe65. The archive structure was readable, but unzip -t requested a password and could not verify/decompress the entry without it. No claim is made about the actual contents of rhino2.jpg.

One carved USB JPEG, 00105864.jpg, is also 230,665 bytes. This is only a size match. Because the ZIP entry could not be decrypted and hashed, the relationship between the two files remains unproven. No password recovery result was supplied in the current findings.

```
unzip -t working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020
```

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ unzip -t working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020

Archive:  working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020
[working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020] rhino2.jpg password:
   skipping: rhino2.jpg              incorrect password
Caution:  zero files tested in working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020.
1 file skipped because of incorrect password.

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
```

zipinfo -v working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ zipinfo -v working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020
Archive:  working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020
There is no zipfile comment.

End-of-central-directory record:
-------------------------------

  Zip archive file size:                  230566 (00000000000384A6h)
  Actual end-cent-dir record offset:      230544 (0000000000038490h)
  Expected end-cent-dir record offset:    230544 (0000000000038490h)
  (based on the length of the central directory and its expected offset)

  This zipfile constitutes the sole disk of a single-part archive; its
  central directory contains 1 entry.
  The central directory is 56 (0000000000000038h) bytes long,
  and its (expected) offset in bytes from the beginning of the zipfile
  is 230488 (0000000000038458h).


Central directory entry #1:
---------------------------

  rhino2.jpg

  offset of local header from start of archive:   0
                                                  (0000000000000000h) bytes
  file system or operating system of origin:      MS-DOS, OS/2 or NT FAT
  version of encoding software:                   2.0
  minimum file system compatibility required:     MS-DOS, OS/2 or NT FAT
  minimum software version required to extract:   2.0
```

sha256sum working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020

```
┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$ sha256sum working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020
b7091a0b4295a561bf76969073ede126a5ca75899bb115447e67272f95d5a29f  working/ftp_recovered/137.030.122.253.02002-137.030.120.040.00020

┌──(kali㉿kali)-[~/icdfaDF/SBT-DF204-Case4]
└─$
```

9. HTTP traffic and rhino.exe

In rhino3.log, frame 110 recorded an HTTP GET request at 2004-04-28 17:16:23.343061 -0400 (21:16:23.343061Z) from 137.30.123.234 to 137.30.120.37 on TCP stream 3 for /~gnome/rhino.exe. Frame 274 recorded a server response at 17:16:24.214222 -0400 (21:16:24.214222Z): HTTP 200 OK, content type application/octet-stream, content length 145,920 bytes.

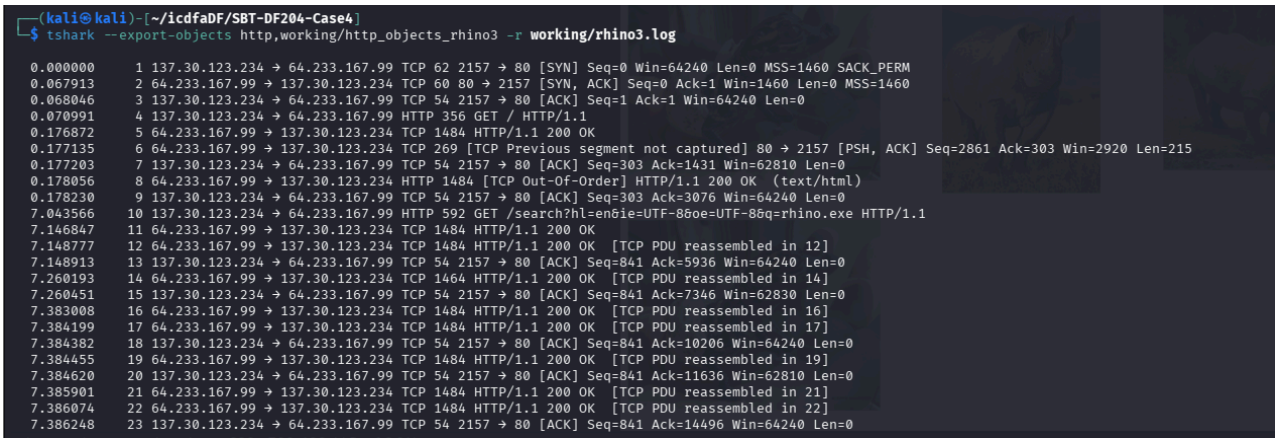
The exported object working/http_objects_rhino3/rhino.exe was 145,920 bytes and identified as a PE32 Windows executable for Intel 80386. Its SHA-256 was 93e70049b60bf5691b43d9e256ca8f459f6bfac832097985bcc6a9040b5ca555. Static inspection recorded imports including DeviceIoControl, DefineDosDeviceW and device-interface/notification

APIs. The embedded PE timestamp was reported as 17 August 2001, but embedded timestamps can be inaccurate and are not independently verified build times. The executable was not executed.

Other HTTP objects were exported, including HTML and GIF resources. A search request containing rhino.exe was also present in the exported objects. A request plus 200 OK and a matching exported object supports that the capture contains a response body of the reported size; it does not establish who intentionally requested it or what the executable would do when run.

tshark -r working/rhino3.log -Y 'http.request' -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e tcp.stream -e http.request.method -e http.host -e http.request.uri > reports/rhino3_http_requests.txt

tshark --export-objects http,working/http_objects_rhino3 -r working/rhino3.log



file working/http_objects_rhino3/rhino.exe

sha256sum working/http_objects_rhino3/rhino.exe

objdump -x working/http_objects_rhino3/rhino.exe > reports/rhino_exe_pe_headers.txt

strings -a -n 5 working/http_objects_rhino3/rhino.exe > reports/rhino_exe_strings.txt

10. Correlated timeline (UTC)

The capture timestamps shown in the available outputs use timezone -0400. The timeline below converts these recorded times to UTC by adding four hours. The capture's clock accuracy was not independently verified.

UTC timestamp	Evidence ID	Artefact / reference	Direct observation	Limit
2004-04-26 22:21:49.954 Z	E05	rhino.log, frame 1546, FTP control stream 69	STOR rhino1.jpg request recorded.	Does not identify the human at the client.
2004-04-26 22:22:08.905 Z	E06	rhino.log, frame 1649, FTP	STOR rhino3.jpg	First transfer used ASCII mode and

		control stream 72	request recorded.	included a server warning.
2004-04-26 22:22:16.091 Z	E07	rhino.log, frame 1763, FTP control stream 72	Second STOR rhino3.jpg request recorded.	Recovered stream hash matches the first upload; intent is unknown.
2004-04-26 22:26:46.671 Z	E08	rhino.log, frame 5647, FTP control stream 305	STOR contraband.zip request recorded.	Archive entry remains encrypted and unverified.
2004-04-28 21:16:23.343 Z	E09	rhino3.log, frame 110, TCP stream 3	GET /~gnome/rhino.exe request recorded.	Request alone does not establish successful retrieval.
2004-04-28 21:16:24.214 Z	E10	rhino3.log, frame 274, TCP stream 3	HTTP 200 OK; application/octet-stream; length 145920.	Does not attribute the request to a named person or prove execution.
Analysis date not original-file time	E11	RHINOUSB.dd / working/usb_carved/jpg/	Seven JPEG candidates carved from unallocated filesystem content.	Carving does not establish original filenames or original creation times.

The timeline contains six network events across FTP and HTTP artefacts, plus the USB carving finding. The USB carving is not assigned a contemporaneous timestamp because no reliable original file timestamp was established.

11. Evidence log

Evidence ID	Tool / method	Source record or output	Finding supported	Screenshot
E01	sha256sum; chmod; file	evidence/rhino.log; reports/original_hashes.txt	Capture identity/integrity record.	S1

E02	sha256sum; chmod; file	evidence/rhino2.log; reports/original_hashes.txt	Capture identity/integrity record.	S1
E03	sha256sum; chmod; file	evidence/rhino3.log; reports/original_hashes.txt	Capture identity/integrity record.	S1
E04	sha256sum; fsstat; fls	evidence/RHINOUSB.dd; working/RHINOUSB.dd	USB image integrity and FAT16 listing.	S1–S2
E05	tshark; tcpflow; file; sha256sum	rhino.log frame 1546; FTP data stream port 1657	rhino1.jpg upload and recovered JPEG hash.	S5–S6
E06	tshark; tcpflow; file; sha256sum	rhino.log frame 1649; FTP data stream port 1660	First rhino3.jpg upload and recovered object.	S5–S6
E07	tshark; tcpflow; sha256sum	rhino.log frame 1763; FTP data stream port 1662	Second rhino3.jpg upload; hash identical to E06.	S5–S6
E08	tshark; tcpflow; zipinfo; unzip -t	rhino.log frame 5647; FTP data stream port 2002	contraband.zip upload; encrypted rhino2.jpg entry.	S5; S7
E09	tshark; HTTP object export	rhino3.log frame 110; TCP stream 3	GET /~gnome/rhino.exe.	S8
E10	tshark; file; sha256sum; objdump	rhino3.log frame 274; working/http_objects_rhino3/rhino.exe	200 OK and static PE identification.	S8–S9
E11	blkls; foremost; exiftool; identify	working/usb_unallocated.bin; working/usb_carved/jpg/	Seven carved JPEG candidates; duplicate pair identified.	S3–S4

12. Limitations and alternative explanations

1. User attribution: IP addresses, FTP account activity, and HTTP requests identify network endpoints or protocol records, not the person at a keyboard.

2. Possession and intent: wildlife imagery, filenames, and a ZIP entry name do not independently prove illegal possession, knowledge, or intent.

3. Encryption: rhino2.jpg was not decrypted or hash-verified. The size match between the ZIP entry's reported uncompressed size and carved file 00105864.jpg is insufficient to establish identity.

4. Deleted and hidden data: the directory-entry search and tsk_recover yielded no deleted files, but carving found candidates in unallocated content. A completed steganography detection/recovery result was not present in the supplied notes, so hidden content is neither confirmed nor ruled out.

5. Timestamp interpretation: packet times were converted from -0400 to UTC, but capture-clock accuracy was not verified. Working-file modification times and embedded PE timestamps are not treated as original creation times.

6. Capture completeness: a packet capture can be partial or omit traffic. A lack of an observed record in these supplied traces does not prove an event never occurred.

7. Duplicate and cross-source comparison: two carved USB JPEGs were duplicates of each other, and the two rhino3.jpg FTP streams were duplicates of each other. No exact hash match was found between the carved USB JPEGs and the recovered FTP JPEGs.

conclusion

The supplied evidence supports a sequence of FTP upload requests on 26 April 2004 and an HTTP request/response for a Windows executable on 28 April 2004. Reconstructed files and hashes provide corroboration for the identified object types and the duplicate rhino3.jpg uploads. The USB image contains two allocated recipe files and seven JPEG candidates carved from unallocated filesystem content. The encrypted archive's entry contents and any steganographic content remain unverified in the available findings. The technical evidence does not establish a named person's actions, original file provenance, or illegal possession beyond reasonable alternative explanations. Further conclusions should be limited to results supported by the saved outputs and labelled screenshots.